CODE-A-NOVA

Full Stack Developer Internship Task

Hackathon Registration Portal

Role: Full Stack Developer Intern

Stack: MERN + Nodemailer

Level: Intermediate

Project Overview

You are required to design and develop a fully functional Hackathon Registration Portal using the MERN Stack (MongoDB, Express.js, React.js, Node.js) with complete CRUD operations. The platform will allow participants to register, verify their identity via OTP, log in to a personal dashboard, and register for a hackathon event with full team management. Confirmation emails must be sent via Nodemailer from a dedicated Code-a-Nova project email account configured through environment variables.

1. Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Tailwind CSS / Material UI	UI components and routing
Backend	Node.js + Express.js	REST API and business logic
Database	MongoDB + Mongoose	Data persistence and schemas

Layer	Technology	Purpose
Auth	JWT + bcrypt	Secure authentication
Email	Nodemailer	OTP and confirmation mails
OTP	Custom 6-digit OTP generator	Email verification

2. Required Components / Pages

#	Component	Description
1	Home	Landing page: hero section, event highlights, timeline, prizes and CTA buttons (Register Now / Login).
2	Navbar	Responsive navigation: Logo (Code-a-Nova), links to Home, About, Contact, Login, Signup. Mobile hamburger menu.
3	Footer	Company footer with copyright, social links, contact email, and quick-nav.
4	Signup	Registration: Full Name, Email, Password, Confirm Password. On submit, send OTP to email and redirect to OTP verification.
5	OTP Verify	6-digit OTP input. Verify OTP, mark account as verified in DB, and redirect to Login page. Include Resend OTP.

#	Component	Description
6	Login	Email + Password login. On success, generate JWT and redirect to Dashboard.
7	Dashboard	Private route. Shows user profile and registered event details. If NOT registered, show a 'Register Now' button linking to the Registration Form.
8	Registration	Full hackathon registration form (see the Form Fields section). Dynamic team member entry. On submit, save to DB and send confirmation email.
9	Contact	Contact form: Name, Email, Message. Display the Code-a-Nova contact email.

3. Hackathon Registration Form Fields

The Registration Page must collect the following fields. All fields are mandatory unless marked optional.

#	Field Label	Input Type	Notes / Validation
1	Full Name	Text Input	Min 3 chars, letters only
2	Email Address	Email Input	Valid email, pre-filled from auth session
3	Mobile Number	Number Input	10-digit Indian mobile number

#	Field Label	Input Type	Notes / Validation
4	College Name	Dropdown / Select	Pre-defined list of colleges
5	Course	Dropdown / Select	B.Tech, M.Tech, BCA, MCA, BSc, etc.
6	Branch	Dropdown / Select	CSE, IT, ECE, ME, CE, etc.
7	Year of Study	Dropdown / Select	1st, 2nd, 3rd, 4th year
8	Domain / Track	Dropdown / Select	AI/ML, Web Dev, IoT, Cyber Security, Blockchain, etc.
9	Project Name	Text Input	Min 5 characters
10	PPT Link	URL Input	Google Drive / OneDrive shareable link
11	Prototype Link	URL Input	GitHub / Figma / live demo link
12	Demo Video Link	URL Input	YouTube / Drive link (Optional)
13	Team Size	Number Select (1-5)	Dynamically renders member entry boxes below

#	Field Label	Input Type	Notes / Validation
14	Team Members	Dynamic Fields	For each member: Member Name + Member Email (based on Team Size)

Dynamic Team Members: When Team Size is selected (e.g., 3), the form dynamically renders 3 entry boxes, each asking for Member Name and Member Email. Team Size = 1 means solo (no extra member boxes).

4. User Flow and Authentication Logic

- **Step 1 – Signup:** User fills the Signup form, the backend creates an unverified account, an OTP is generated and emailed via Nodemailer, and the user is redirected to the OTP Verification page.
- **Step 2 – OTP Verify:** User enters the 6-digit OTP, the backend verifies it (time-limited: 10 minutes), marks the user as verified in MongoDB, and redirects to the Login page.
- **Step 3 – Login:** User enters Email + Password, bcrypt.compare() is run, a JWT is issued and stored in localStorage or an httpOnly cookie, and the user is redirected to the Dashboard.
- **Step 4 – Dashboard:** Protected route (JWT guard). Displays the user profile. If no hackathon registration is found, show the 'Register Now' CTA.

- **Step 5 – Registration:** User fills the Hackathon Registration Form. On submit, save to DB, send a confirmation email with event and team details via Nodemailer, and show a success screen.
- **Step 6 – Logout:** Clear the JWT token from storage and redirect to Home.

5. Nodemailer Email Configuration

Email	Trigger Event	Email Content
Project email (set in .env)	User Signup (OTP)	6-digit OTP with 10-minute expiry. Sender name: Code-a-Nova
Project email (set in .env)	Hackathon Registration	Confirmation with participant name, event, team details, domain, project name

Important: Do NOT include any candidate name or personal sender name in emails. Always use Code-a-Nova as the sender name. Enable Gmail App Passwords for SMTP authentication in Nodemailer if a Gmail account is used.

6. CRUD Operations – API Endpoints

Operation	Endpoint	Description
CREATE – Register	POST /api/auth/signup	Create new user account

Operation	Endpoint	Description
CREATE – OTP Verify	POST /api/auth/verify-otp	Verify OTP, activate account
READ – Login	POST /api/auth/login	Authenticate user, return JWT
READ – Profile	GET /api/user/me	Fetch logged-in user's profile (protected)
CREATE – Hackathon	POST /api/hackathon/register	Submit hackathon registration form
READ – My Entry	GET /api/hackathon/my-entry	Fetch user's hackathon registration
UPDATE – Profile	PUT /api/user/update	Update user details (optional)
DELETE – Entry	DELETE /api/hackathon/:id	Delete registration (admin / optional)

7. Deliverables

- **GitHub Repository:** Public repo with /client (React) and /server (Node/Express) folders. Include a detailed README.md.
- **Live Deployment:** Frontend on Vercel/Netlify and backend on Render/Railway. Share both live URLs.

- **Environment Variables:** Use .env for all secrets (MONGO_URI, JWT_SECRET, EMAIL_PASS). Include a .env.example in the repo.
- **README.md:** Must include: project overview, tech stack, setup steps, API docs, .env guide, and screenshots.
- **Demo Video:** A 3–5 minute screen recording covering: signup, OTP, login, dashboard, registration, and confirmation email.

8. Evaluation Criteria

Criteria	Marks
Functional Completeness (all flows working end-to-end)	30
Code Quality, Structure and Readability	20
UI/UX Design and Responsiveness (mobile + desktop)	20
Authentication and Security (JWT, bcrypt, OTP expiry)	15
Nodemailer Integration (OTP email + Confirmation email)	10
README, Deployment and Demo Video	5
Total	100

9. Important Rules and Guidelines

- Do NOT include any candidate name or personal information in email templates.

- Always use Code-a-Nova as the sender name in all emails.
- All routes except Home, Login, and Signup must be protected routes (JWT middleware guard).
- Passwords must be hashed using bcrypt before storing in MongoDB. Never store plain-text passwords.
- OTPs must be time-limited (expire after 10 minutes) and invalidated after first use.
- Use .env for all secrets. Never hardcode credentials or API keys in source code.
- Code must be well-commented and follow consistent naming conventions (camelCase).
- The application must be fully responsive, tested on mobile, tablet, and desktop.
- Plagiarism or code copied from others will result in immediate disqualification.